

Data/Math 363 Introduction to Statistical Methods

T/TH 9:30 – 10:45 AM M Pacheco ILC, Rm 119

Description of Course

Data/Math 363 will be using your background in the natural or social sciences, the humanities, or engineering and your previous knowledge of algebra, calculus, and linear algebra to consider the issues of collection, model derivation and analysis, interpretation, explanation, and presentation of data. The objective of this course is to take advantage of the coherent body of knowledge provided by statistical theory, having an eye consistently on the application of the subject. This approach will allow you to extend your ability to use methods in data science beyond those given in the course.

Course Prerequisites or Co-requisites

Math 223 and CSC 110 or ISTA 130 and have completed Math 310, 313, or 215, or are registered for Math 310 or 313.

Instructor and Contact Information

Instructor: Anisha Mathias, Joseph Watkins

Email: anishajmathias@arizona.edu

Office: MATH-715, ENR2 S321

Phone: 520-621-6892

Office Hours: Posted in D2L

D2L Link: <https://d2l.arizona.edu/d2l/home/1712440>

Course Format and Teaching Methods

Math 363 is a 3-credit hour course in the in-person modality. In this class mode, the students and instructors attend class in person. Access to all course materials such as readings, reading assignments, websites, or videos is provided in the D2L course site.

This course is scheduled to meet Tuesdays and Thursdays from 9:30 am – 10:45 am in M Pacheco ILC, Rm 119. It is expected that all students will attend class regularly and be active participants.

This course is taught using a flipped format. Thus, you will typically be watching 5 to 6 short lectures, answering 2 to 3 checkpoint exercises, and submitting responses at the beginning of class. The class will begin with a discussion to assess your understanding of the lectures and to discuss the group assignment. Following the group assignment, most of the class will be devoted to working problems in a worksheet having one or two statistics or probability problems. You will need to bring a laptop to class.

Much of the work can be completed during class time and you are encouraged to work together during class. Your homework in this class is to watch the lecture videos.

For lecture recordings, students will access content in D2L. Students may not modify content or re-use content for any purpose other than personal educational reasons. Students accessing unauthorized recordings or using them in a manner inconsistent with UArizona values and educational policies are subject to suspension. All recordings are the property of the faculty member.

Course Objectives

- Organizing data numerically and visually
- Axioms of probability, conditional probability, and independence
- Random variables and expectation with emphasis on parametric families
- Law of Large Numbers and Central Limit Theorem
- Estimation, bias and variance, confidence intervals
- Hypothesis testing, significance, and power
- Likelihood ratio tests such as proportion tests, t-tests, chi-square tests, and analysis of variance

Expected Learning Outcomes

Upon completion of this course, students should be able to:

1. Understand what statistical thinking is
2. Integrate statistical thinking with scientific procedures and quantitative modeling
3. Ask statistics experts productive questions
4. Implement their ideas using statistical software and other computational tools

Absence and Class Participation Policy

Participating in the course and attending lectures and other course events are vital to the learning process. As such, attendance is required at all lectures and discussion section meetings. Absences may affect a student's final course grade. If you anticipate being absent, are unexpectedly absent, or are unable to participate in class activities, please contact me as soon as possible. The UA's policy concerning Class Attendance and Participation is available at: <https://catalog.arizona.edu/policy/class-attendance-and-participation>

- To request a disability-related accommodation to this attendance policy, please contact the Disability Resource Center at (520) 621-3268 or drc-info@email.arizona.edu.
- If you are experiencing unexpected barriers to your success in your courses, the Dean of Students Office is a central support resource for all students and may be helpful. The Dean of Students Office is located in the Robert L. Nugent Building, room 100, or call 520-621-7057.
- The UA policy regarding absences for any sincerely held religious belief, observance or practice will be accommodated where reasonable, <http://policy.arizona.edu/human-resources/religious-accommodation-policy>.
- Students who need to miss more than 1 week of classes (or the equivalent for accelerated courses) should provide documentation to the Dean of Students office (DOS-deanofstudents@email.arizona.edu). Documentation should be submitted prior to a planned absence. Documentation may be submitted during or after an unplanned absence. It is the student's responsibility to notify the instructor in advance of an absence related to religious

observation or an activity for which a Dean's excuse has been granted, and to arrange for how any missed work will be handled. See <https://catalog.arizona.edu/policy/class-attendance-and-participation> for more information.

Makeup Policy for Students Who Register Late

Students who register after the first day of classes are responsible for notifying the instructor of the course by the beginning of the next class meeting after they have registered. Makeup work will be provided at the discretion of the instructor.

Course Communications

Announcements and important course information may be sent out via official University email or through D2L. It is the student's responsibility to check for messages and announcements regularly.

Required Texts or Readings

An Introduction to the Science of Statistics: From Theory to Implementation by Joseph C. Watkins (the PDF of the textbook is provided for free, [click here](#))

Required Materials

- Computer: For this class you will need daily access to a laptop or web-enabled device, and regular access to reliable internet signal. Our course work will take place in D2L and Gradescope. Note: Enrolled students can borrow technology from the UA Library on a first come, first served basis. See <https://new.library.arizona.edu/tech/borrow-for-details>.
- Calculator: A graphing calculator; any model in TI-83 or TI-84 series is recommended. Please see [this list](#) for calculators not allowed by the math department.
- R or RStudio: [R](#) is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To download R, please choose your preferred CRAN mirror (It doesn't matter which you choose - preferably choose one in a country where you will most be using R for this course and in the future). As with any computer software, the syntax in R will seem awkward at first. Many of you will also want to download [RStudio](#), which is also free. RStudio provides a graphical user interface that will make the use of R go more smoothly.
- Gradescope: You will receive an email from the application Gradescope that contains instruction on how to login and create a password. Please sign-up for Gradescope as soon as you receive the email. This is the platform where you will submit your written homework assignments each week.

Assignments and Examinations: Schedule/Due Dates

There are 4 components to homework:

- Checkpoints (5% of course grade) – After watching the lecture videos, you will complete 2-3 checkpoint problems to check for understanding of the lecture. The location of the checkpoint prompts is in the checkpoint document provided in the before class section in D2L for each class period. Checkpoints must be completed on paper and brought to each class period. During the presentations, it is your responsibility to annotate on your provided solutions. At the beginning of each class, students will volunteer to present their solution to the checkpoint problems. At the end of the semester, an additional checkpoint grade will be awarded for the presentation during class time. The grading scheme for the checkpoint presentations can be found in D2L. Each checkpoint will be worth 4 points and will be submitted to the instructor during class time. No late checkpoints will be accepted. The lowest 2 checkpoints will be dropped, and the remaining assignments will be averaged and scaled to 40 points in the course.
- Group Activities (10% of course grade) – During non-exam classes, students will complete a group activity via the whiteboards in the classroom. Each group activity will be worth 10 points and will be submitted as a group to Gradescope by 11:00 am the same day. No late group activities will be accepted. Please note you must be present in class and working with a group to complete and receive credit for the activity. The lowest 2 group activities will be dropped, and the remaining assignments will be averaged and scaled to 40 points in the course. Due to the number of group assignments that will be dropped, no makeup assignments will be given without a Dean of Students excuse.
- Worksheets (15% of course grade) – Written homework will be assigned weekly with time to work on it during each class. Solutions should show all work with proper notation and often will include a corresponding file containing all appropriate R code. Each assignment will be generally 1-2 problems with several parts that are designed to deepen and integrate your knowledge of that week's topics. You are encouraged to work with others to complete the worksheets. Each worksheet will be worth 20 points and will be submitted individually to Gradescope by 11:59 PM (Arizona time) on Thursdays, unless otherwise stated. Late worksheets will be accepted up to 24 hours after the due date for up to 85% of the credit. The lowest 1 worksheet will be dropped, and the remaining assignments will be averaged and scaled to 40 points in the course.
- Project (15% of course grade) – Students will design and complete a project that analyzes data using statistical software. See guidelines posted on the D2L course page. The project grade will consist of 5 graded submissions to Gradescope to be submitted with your project partner(s).
 - Project Proposal
 - Introduction and Methods
 - Preliminary Report
 - Presentation
 - Final Report

There are two midterm exams worth 30% of your overall grade (15% for each exam). The dates, times, and content for the midterms are given below. All exams will be proctored and given in class time.

- Exam 1 – Material covered through Mass/Density Functions
 - Thursday, March 5, 2026
- Exam 2 – Material covered from Law of Large Numbers through Hypothesis Testing

- Thursday, April 23, 2026

All grading disputes must be addressed within one week of an assignment being returned.

Final Examination

A comprehensive Final Exam will be given in class on Tuesday, May 12, 2026 from 8:00 am – 10:00 am. Please put this date in your calendar immediately. A study guide for the final exam will be posted as a PDF document in D2L.

The University final exam schedule and rules relating to final examinations may be found at: <https://registrar.arizona.edu/faculty-staff-resources/room-class-scheduling/schedule-classes/final-exams>

Grading Scale and Policies

Grades	Points	Percent of Grade	You are Guaranteed a Grade of:	
Checkpoints	40	5%	A	if you earn at least 360 points (90%)
Worksheets	40	15%	B	if you earn at least 320 points (80%)
Group Activities	40	10%	C	if you earn at least 280 points (70%)
Project	60	15%	D	if you earn at least 240 points (60%)
Exams	120	30%		
Final Exam	100	25%		
Total possible points	400	(100 %)		

Please note that neither exam scores nor final grades will be curved. No extra credit or bonus points are offered in this course.

Incomplete (I) or Withdrawal (W):

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, which are available at <http://catalog.arizona.edu/policy/grades-and-grading-system#incomplete> and <http://catalog.arizona.edu/policy/grades-and-grading-system#Withdrawal> respectively.

A student may withdraw from the course with deletion from record through January 27th, 2026, using UAccess. A student may withdraw with a grade of “W” through March 31st, 2026. It is suggested that students consult their academic advisor before withdrawal from any course.

Scheduled Topics/Activities

Week	Topics	Assignments Due
1	Displaying Data	Checkpoints; Group Activity
2	Describing Data, Correlation and Regression	Checkpoints; Group Activity; Worksheets
3	Correlation, Regression and Producing Data	Checkpoints; Group Activity; Worksheets
4	Probability, Project Work	Checkpoints; Group Activity; Worksheets; Project Proposal
5	Probability, Random Variables	Checkpoints; Group Activity; Worksheets
6	Random Variables, Expected Value	Checkpoints; Group Activity; Worksheets
7	Expected Value, Mass/Density Functions	Project Intro and Methods; Checkpoints; Group Activity
8	Exam, Law of Large Numbers	Checkpoints; Group Activity; Worksheets
9	Central Limit Theorem, Estimation	Checkpoints; Group Activity; Worksheets
10	Estimation	Checkpoints; Group Activity; Worksheets
11	Estimation	Checkpoints; Group Activity; Worksheets
12	Hypothesis Tests	Checkpoints; Group Activity; Worksheets
13	Hypothesis Tests, Project Work	Checkpoints; Group Activity; Worksheets; Project Preliminary Report
14	Exam 2	Worksheet
15	t procedures; Goodness of Fit	Checkpoints; Group Activity; Worksheets
16	ANOVA	Final Project Report; Project Presentation

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, reading a newspaper, making phone calls, web surfing, etc.).

Threatening Behavior Policy

The UA Threatening Behavior by Students Policy prohibits threats of physical harm to any member of the University community, including to oneself. See <http://policy.arizona.edu/education-and-student-affairs/threatening-behavior-students>.

Accessibility and Accommodations

At the University of Arizona, we strive to make learning experiences as accessible as possible. If you anticipate or experience barriers based on disability or pregnancy, please contact the Disability Resource Center (520-621-3268, <https://drc.arizona.edu/>) to establish reasonable accommodations.

Code of Academic Integrity

Students are encouraged to share intellectual views and discuss freely the principles and applications of course materials. However, graded work/exercises must be the product of independent effort unless otherwise instructed. Students are expected to adhere to the UA Code of Academic Integrity as described in the UA General Catalog. See: <http://deanofstudents.arizona.edu/academic-integrity/students/academic-integrity>.

The University Libraries have some excellent tips for avoiding plagiarism, available at <http://new.library.arizona.edu/research/citing/plagiarism>.

Selling class notes and/or other course materials to other students or to a third party for resale is not permitted without the instructor's express written consent. Violations to this and other course rules are subject to the Code of Academic Integrity and may result in course sanctions. Additionally, students who use D2L or UA e-mail to sell or buy these copyrighted materials are subject to Code of Conduct Violations for misuse of student e-mail addresses. This conduct may also constitute copyright infringement.

Generative AI

In this course any and all uses of generative artificial intelligence (AI)/large language model tools such as ChatGPT, Dall-e, Google Bard, Microsoft Bing, etc. may be considered a violation of the Code of Academic Integrity, specifically the prohibition against submitting work that is not your own. This applies to all assessments in the course, including written assignments, discussions, quizzes, exams, and problem sets. This course policy is driven by the learning goals and desired learning outcomes for the course. The following actions are prohibited:

- Entering all or any part of an assignment statement or test questions as part of a prompt to a large language model AI tool
- Incorporating any part of an AI-written response in an assignment
- Using AI to summarize or contextualize reading assignments or source materials
- Submitting your own work for this class to a large language model AI tool for iteration or improvement.

Nondiscrimination and Anti-harassment Policy

The University of Arizona is committed to creating and maintaining an environment free of discrimination. In support of this commitment, the University prohibits discrimination, including harassment and retaliation, based on a protected classification, including race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, or genetic information. For more information,

including how to report a concern, please see <http://policy.arizona.edu/human-resources/nondiscrimination-and-anti-harassment-policy>

Our classroom is a place where everyone is encouraged to express well-formed opinions and their reasons for those opinions. We also want to create a tolerant and open environment where such opinions can be expressed without resorting to bullying or discrimination of others.

Additional Resources for Students

UA Academic policies and procedures are available at <http://catalog.arizona.edu/policies>

Campus Health

<http://www.health.arizona.edu/>

Campus Health provides quality medical and mental health care services through virtual and in-person care.

Phone: 520-621-9202

Counseling and Psych Services (CAPS)

<https://health.arizona.edu/counseling-psych-services>

CAPS provides mental health care, including short-term counseling services.

Phone: 520-621-3334

The Dean of Students Office's Student Assistance Program

<http://deanofstudents.arizona.edu/student-assistance/students/student-assistance>

Student Assistance helps students manage crises, life traumas, and other barriers that impede success. The staff addresses the needs of students who experience issues related to social adjustment, academic challenges, psychological health, physical health, victimization, and relationship issues, through a variety of interventions, referrals, and follow up services.

Email: DOS-deanofstudents@email.arizona.edu

Phone: 520-621-7057

Survivor Advocacy Program

<https://survivoradvocacy.arizona.edu/>

The Survivor Advocacy Program provides confidential support and advocacy services to student survivors of sexual and gender-based violence. The Program can also advise students about relevant non-UA resources available within the local community for support.

Email: survivoradvocacy@email.arizona.edu

Phone: 520-621-5767

Campus Pantry

Any student who has difficulty affording groceries or accessing sufficient food to eat every day, or who lacks a safe and stable place to live and believes this may affect their performance in the course, is urged to contact the Dean of Students for support. In addition, the University of Arizona Campus Pantry is open for students to receive supplemental groceries at no cost. Please see their website at: campuspantry.arizona.edu for open times.

Furthermore, please notify me if you are comfortable in doing so. This will enable me to provide any resources that I may possess.

Preferred Name & Pronoun

This course affirms people of all gender expressions and gender identities. If you prefer to be called a different name than what is on the class roster, please let me know. Feel free to correct instructors on your preferred gender pronoun. If you have any questions or concerns, please do not hesitate to contact me directly in class or via email (instructor email). If you wish to change your preferred name or pronoun in the UAccess system, please use the following guidelines:

Preferred name: University of Arizona students may choose to identify themselves within the University community using a preferred first name that differs from their official/legal name. A student's preferred name will appear instead of the person's official/legal first name in select University-related systems and documents, provided that the name is not being used for the purpose of misrepresentation. Students are able to update their preferred names in UAccess.

Pronouns: Students may designate pronouns they use to identify themselves. Instructors and staff are encouraged to use pronouns for people that they use for themselves as a sign of respect and inclusion. Students are able to update and edit their pronouns in UAccess.

More information on updating your preferred name and pronouns is available on the Office of the Registrar site at <https://www.registrar.arizona.edu/>.

Safety on Campus and in the Classroom

For a list of emergency procedures for all types of incidents, please visit the website of the Critical Incident Response Team (CIRT): <https://cirt.arizona.edu/case-emergency/overview>

Also watch the video available at https://arizona.sabacloud.com/Saba/Web_spf/NA7P1PRD161/common/learningeventdetail/crtfy0000000000003560

Confidentiality of Student Records

<http://www.registrar.arizona.edu/ferpa>

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.